

Oswin Franco Cervantes

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Statement of Purpose

My goal is to use first-principles physics, electromagnetic modeling, circuit and system-level design, measurement, and computational methods to develop integrated hardware spanning semiconductor circuits, RF, photonics, and quantum systems.

Education

Massachusetts Institute of Technology, Cambridge, MA

Expected May 2028

B.S. Candidate in Physics and Electrical Engineering with Computing

Sept. 2024–Present

Selected coursework: **Graduate:** High-Frequency Integrated Circuits, Electromagnetics; **Coursera:** Microwave Engineering and Antennas, RF and Millimeter-Wave Circuit Design; **EE/Physics:** Solid-State Circuits, Digital Systems Laboratory (Verilog), Semiconductor Electronic Circuits, Signal Processing, Quantum Physics I, Statistical Mechanics; **Computer Science:** Mathematics for Computer Science, Computational Structures, Low-Level Programming

Georgia Tech, Atlanta, GA — Dual Enrollment, Mathematics/Computer Science; GPA: 4.0/4.0

Aug. 2022–May 2024

Integrated Circuit Design Project

22 nm FinFET Amplifier and Control Chip

Spring 2026

MIT Semiconductor Electronic Circuits

Cambridge, MA

- Designed and simulated a 22 nm FinFET CMOS chip combining an analog amplifier, activation switches, and digital control logic; applied semiconductor device physics and large- and small-signal models.
- Created the chip layout in Cadence for a course tapeout and ran Calibre DRC/LVS checks to verify fabrication rules and consistency with the schematic.
- Verified through simulation that the amplifier exceeded targets of 26 dB gain at 1 MHz and 300 mV peak-to-peak output swing while operating 48% below its 500 μ W power limit.

Experience

Undergraduate Researcher — Prof. William D. Oliver

Sept. 2025–Present

MIT Research Laboratory of Electronics, Engineering Quantum Systems (EQuS) Group

Cambridge, MA

- Own the design and testing of niobium coplanar-waveguide filters for hybrid semiconductor quantum-dot and superconducting devices, minimizing crosstalk, passband insertion loss, and out-of-band noise from room temperature through cryogenic operation.
- Create RF layouts in KLayout; model electromagnetic fields in Ansys HFSS/Maxwell using eigenmode and S-parameter analyses plus waveguide, transmission-line, and impedance-matching theory; automate with PyAEDT and extract parasitics with Q3D.
- Collaboratively develop CDES, a Python and gdstk framework for superconducting circuit design; contribute reusable components and fabrication GDS layouts through GitHub version control; fabricate niobium devices on silicon by cleanroom photolithography.

Avionics Engineer

Jan. 2026–Present

MIT Rocket Team

Cambridge, MA

- Design RF circuits and PCB interconnects in Altium; analyze interconnect behavior with HFSS Driven Terminal and Ansys SIwave; compare simulated S-parameters with vector network analyzer measurements.

Hardware Development Engineer (Systems Engineering) Intern

May 2025–Aug. 2025

Amazon Lab126, Advanced Products

Sunnyvale, CA

- Translated ambiguous product requirements with a cross-functional team into system architecture, functional block diagrams, schematics, and PCB layout for a multi-IC solar-charging module compatible with Amazon Kindle devices.
- Designed and simulated buck, boost, and buck-boost power circuits in OrCAD Capture and SPICE and laid out PCBs in Cadence Allegro; measured efficiency with oscilloscopes and electronic loads and investigated MPPT.
- Implemented and debugged I²C for IC integration; assembled and tested 0402, 0603, and 0804 SMD prototypes and debug boards to validate end-to-end functionality and performance; trained new hires on PCB tools.

Technical Skills

Design and Analysis: Electromagnetic modeling, RF/microwave circuits, solid-state circuits, analog/mixed-signal ICs, digital/RTL design, power circuits, S-parameters, transmission lines, impedance matching, SPICE, PCB/system design

Fabrication and Test: 22 nm FinFET custom layout, DRC/LVS, GDS/KLayout, niobium-on-silicon photolithography, cleanroom processing, SMD, VNA and oscilloscope measurements

Software/HDL: Python, C, Java, Julia, MATLAB, Bash, Verilog, VHDL, Bluespec, RISC-V asm.; NumPy, pandas, gdstk, Git

EDA Tools: Ansys (HFSS, Maxwell, Q3D, SIwave, PyAEDT); Cadence (custom IC, AWR Microwave Office, Allegro); Calibre, Altium, OrCAD, KLayout

Honors and Certification

Honors: QuestBridge Scholarship; Amazon Scholar; Jack Kent Cooke Foundation Semifinalist; Stanford Intrologic Scholar

Certification: TestOut Network Pro Certification